

Quantum Computation and Information

Homework Set 3

1. Given two vectors $x, y \in \mathbb{R}^n$, using amplitude encoding we can encode them as quantum states

$$|x\rangle = \sum_{k=1}^n \frac{x_k}{\|x\|} |k\rangle, \quad |y\rangle = \sum_{k=1}^n \frac{y_k}{\|y\|} |k\rangle,$$

where $\|\cdot\|$ denotes Euclidean norm and $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ are orthonormal quantum states.

- (a). Show that the inner product $x^T y = \|x\| \|y\| \langle x|y\rangle$.

Solutions:

$$\begin{aligned} \|x\| \|y\| \langle x|y\rangle &= \|x\| \|y\| \left(\sum_{k=1}^n \frac{x_k}{\|x\|} |k\rangle \right)^\dagger \left(\sum_{k'=1}^n \frac{y_{k'}}{\|y\|} |k'\rangle \right) \\ &= \sum_{k=1}^n \sum_{k'=1}^n x_k y_{k'} \langle k|k'\rangle = \sum_{k=1}^n \sum_{k'=1}^n x_k y_{k'} \delta_{kk'} = \sum_{k=1}^n x_k y_k = x^T y \end{aligned}$$

- (b). The outer product xy^T is an n -by- n matrix. How to encode xy^T quantumly using $|x\rangle$ and $|y\rangle$?

Solutions: $\|x\| \|y\| |x\rangle \langle y|$.

- (c). Suppose the answer to (b) is A . Show that the inner product $x^T y = \text{tr } A$.

Solutions: $A = \|x\| \|y\| \left(\sum_{k=1}^n \frac{x_k}{\|x\|} |k\rangle \right) \left(\sum_{k'=1}^n \frac{y_{k'}}{\|y\|} |k'\rangle \right)^\dagger = \sum_{k=1}^n \sum_{k'=1}^n x_k y_{k'} |k\rangle \langle k'|$. Thus

$$\begin{aligned} \text{tr } A &= \sum_{\bar{k}=1}^n \langle \bar{k} | A | \bar{k} \rangle = \sum_{\bar{k}=1}^n \sum_{k=1}^n \sum_{k'=1}^n x_k y_{k'} \langle \bar{k} | k \rangle \langle k' | \bar{k} \rangle \\ &= \sum_{\bar{k}=1}^n \sum_{k=1}^n \sum_{k'=1}^n x_k y_{k'} \delta_{\bar{k}k} \delta_{\bar{k}k'} = \sum_{\bar{k}=1}^n x_{\bar{k}} y_{\bar{k}} = x^T y. \end{aligned}$$

- (d). Design a quantum algorithm to determine the sign of $x^T y$, i.e., if it is positive or negative.

Solutions: First, prepare quantum state $|\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle |x\rangle + |1\rangle |y\rangle)$, where $\{|0\rangle, |1\rangle\}$ are usual computational basis of a ancillary qubit. Second, measure the ancillary qubit using projection measurement onto state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Note that $|\phi\rangle = \frac{1}{2}|+\rangle(|x\rangle + |y\rangle) + \frac{1}{2}|-\rangle(|x\rangle - |y\rangle)$, Thus, the measurement has an success probability $P = (\frac{1}{2}(\langle x| + \langle y|)(\frac{1}{2}(|x\rangle + |y\rangle)) = \frac{1}{2}(1 + \langle x|y\rangle)$. Given (a), if $P > 1/2$, the sign of $x^T y$ is +; and if $P < 1/2$, the sign of $x^T y$ is -.

2. Given N vectors $x_1, x_2, \dots, x_N \in \mathbb{R}^n$. Define an N -by- N matrix K with entry $K_{ij} = x_i^T x_j$. Matrix K can be encoded as a quantum state described by density matrix $\rho = \sum_{i=1}^N \sum_{j=1}^N K_{ij} |i\rangle \langle j|$, where $\{|1\rangle, |2\rangle, \dots, |N\rangle\}$ are orthonormal quantum states. Explain in detail how you would prepare quantum state ρ .

Solutions: Pls see the 3rd paragraph of page 2 of reference "Quantum Support Vector Machine for Big Data Classification" (ML10 of Reading Materials on the course Canvas). Most of Problem 1 was designed to help understand it.